

IMEN JARRAYA

Postdoctoral Fellow RIOTU Lab

CONTACT

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Location

Riyadh, Saudi Arabia

EDUCATION

Ph.D. Electrical Eng.

ENIS & ESTACA 2016–2021

M.Sc. Electrical

ENIS, 2008–2010

TEACHING EXPERIENCE

2023–Present

Research Mentoring, PSU

2021–2023

Lecturer, ISET

2010–2015

Lecturer, ISGIS & Bio Institute

COURSES TAUGHT

- Real-Time Systems
- Embedded Systems
- Internet of Things
- Power Electronics
- Industrial Computing
- Automation
- Biomedical Measurement

LANGUAGES

Arabic Native

French Fluent

English Professional

TEACHING STATEMENT

Imen Jarraya, Ph.D.

TEACHING PHILOSOPHY

"The goal of education is not to fill a bucket, but to light a fire." — W.B. Yeats

My teaching philosophy is grounded in the conviction that engineering education must bridge rigorous theoretical foundations with hands-on, real-world application. As an Electrical Engineer, University Lecturer, and Postdoctoral Fellow at Prince Sultan University within the Robotics and Internet of Things Unit (RIOTU), I have witnessed how active learning, project-based pedagogy, and research-informed teaching help students become independent and creative problem-solvers.

TEACHING EXPERIENCE (8+ YEARS)

2023–Present: Research Mentor at Prince Sultan University (RIOTU Lab). Mentoring students and junior researchers in UAV navigation, AI, embedded systems, robotics, and IoT.

2021–2023: University Lecturer at ISET, Tunisia. Delivered lectures and labs in Real-Time Systems, Embedded Systems Programming, and IoT. Designed course materials and supervised end-of-studies projects.

2010–2015: University Lecturer at ISGIS and Institute of Biotechnology, Tunisia. Taught Industrial Computing, Power Electronics, Automation, Electrical Engineering, and Biomedical Measurement. Designed laboratory exercises and mentored students.

PEDAGOGICAL APPROACHES

Active Learning & Hands-On Instruction: I design courses around active learning using real hardware (Arduino, STM32F4, ESP32, Raspberry Pi, Jetson Nano), reinforcing theory through tangible outcomes.

Project-Based Learning: I have supervised 10+ graduation projects in IoT, robotics, power systems, and photovoltaics, building student ownership and professional engineering skills.

Research-Integrated Teaching: I connect teaching to my research in UAV navigation, battery monitoring, hybrid energy management, embedded systems, and IoT, exposing students to frontier problems.

Inclusive & Adaptive Teaching: I adapt to diverse student backgrounds using visual aids, simulations (MATLAB/Simulink), hardware demonstrations, and structured laboratory activities.

MENTORING & SUPERVISION

- Ph.D. thesis co-supervision (ENIS Sfax, from 2019) on EV routing optimization
- 10+ graduation projects supervised and evaluated
- Current mentoring at PSU in UAV navigation, AI, embedded systems, robotics, IoT
- Committed to promoting women in STEM (recognized at 5th International Day of Women and Girls in Science, AIDA Lab, PSU 2025)

FUTURE TEACHING VISION

I aspire to develop graduate-level courses in: Autonomous Systems & UAV Navigation, AI for Energy Systems, Embedded AI & Edge Computing, and Robotics & Internet of Things. I envision laboratory environments where students work on real engineering and research problems with genuine societal impact aligned with Saudi Vision 2030.